CLOUD COMPUTING LAB 6

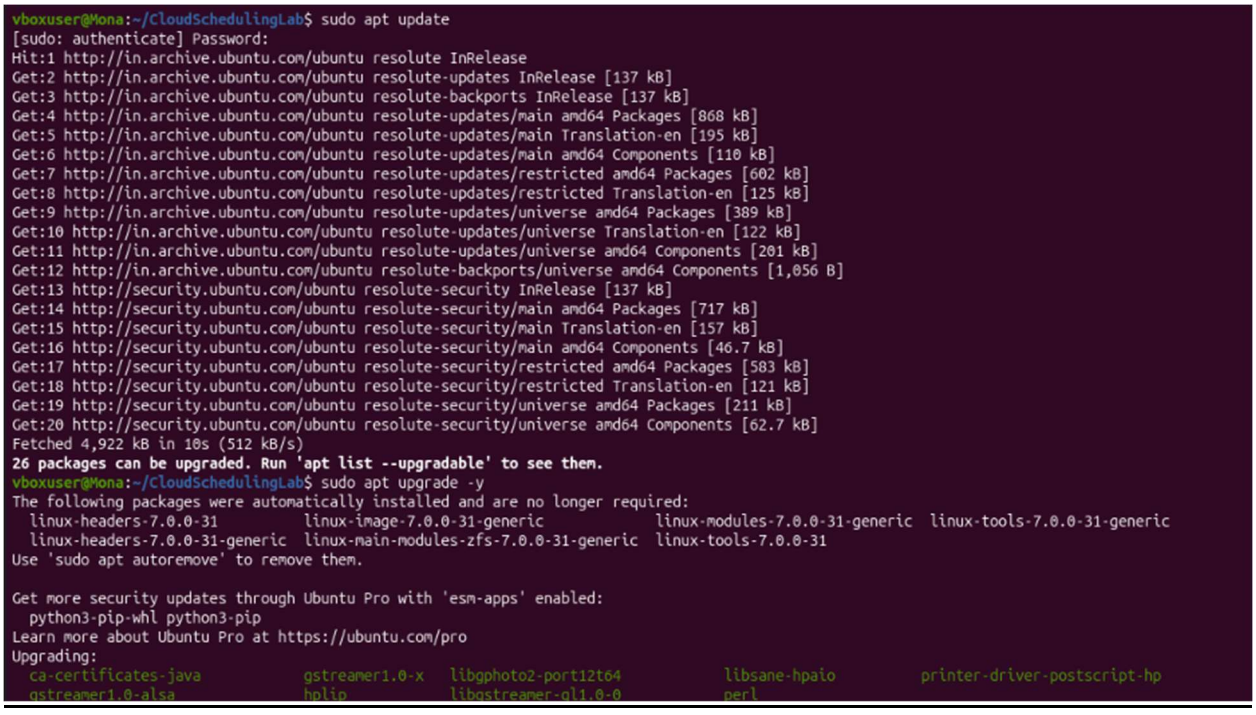
Task 1: VM Setup :

Step 1: Create a Virtual Machine,

Step 2: Install Ubuntu,

Step 3: Update Ubuntu,

Step 4: Install Basic Utilities



Task 2A: Introduction to AWS Management Console What is AWS?

Amazon Web Services (AWS) is a cloud computing platform that provides services such as computing, storage, databases, networking, security and monitoring. AWS allows users to create and manage cloud resources without maintaining physical infrastructure.

Step 1: Access AWS Management Console:

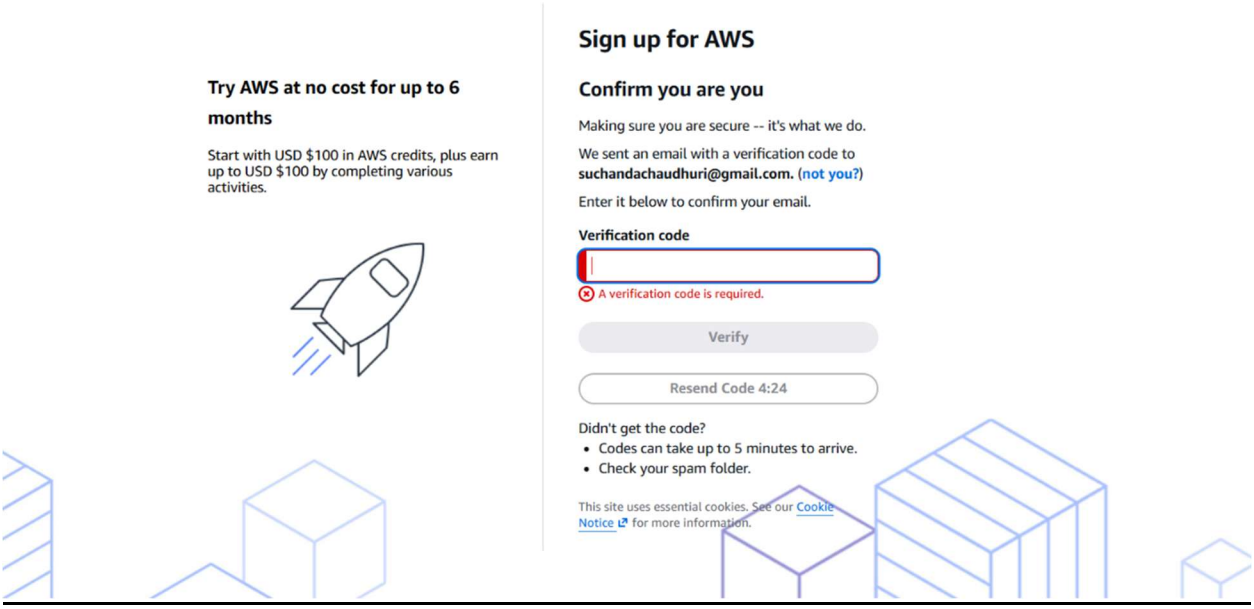
1. Open a web browser.

2. Visit https://console.aws.amazon.com/

3. Sign in using an authorised AWS account.

4. Explore the AWS Management Console home page.

5. Observe the search bar used to locate different AWS services.



Task 2B: Implementation of Cloud Storage Using Amazon S3

Introduction to Amazon S3

Amazon Simple Storage Service (Amazon S3) is an object storage service used to store and retrieve data over the internet.

Step 1: Open Amazon S3

1. Log in to the AWS Management Console.

2. Search for S3.

3. Open Amazon S3.

4. Select Create bucket.

Step 2: Create an S3 Bucket

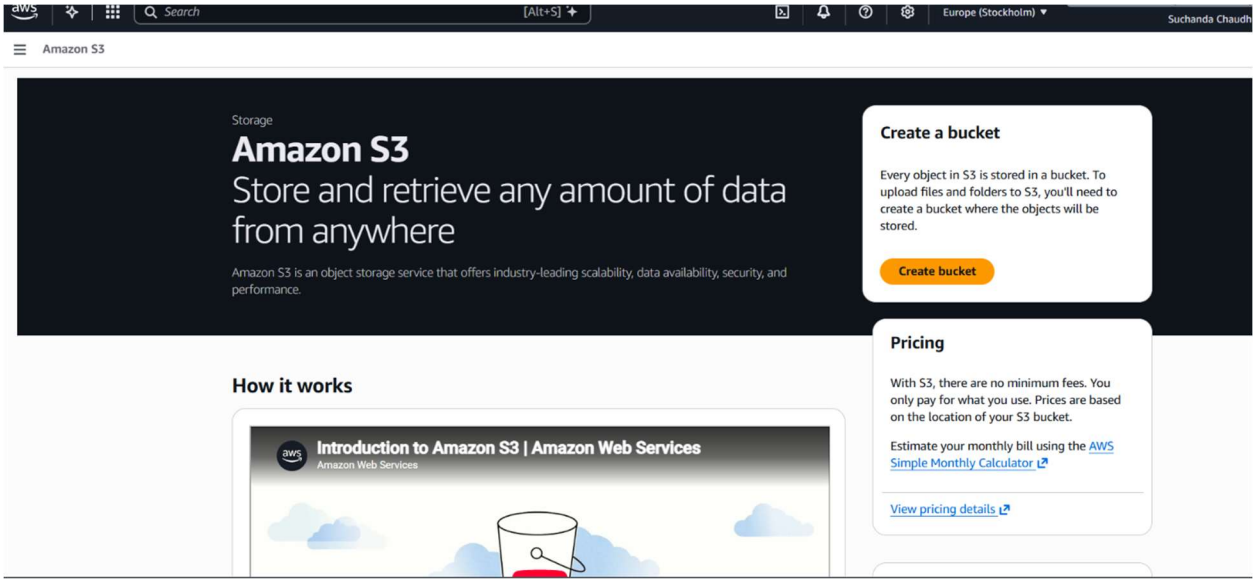
1. **Enter a globally unique bucket name.**
   Example: cse12045-cloud-lab6-student01

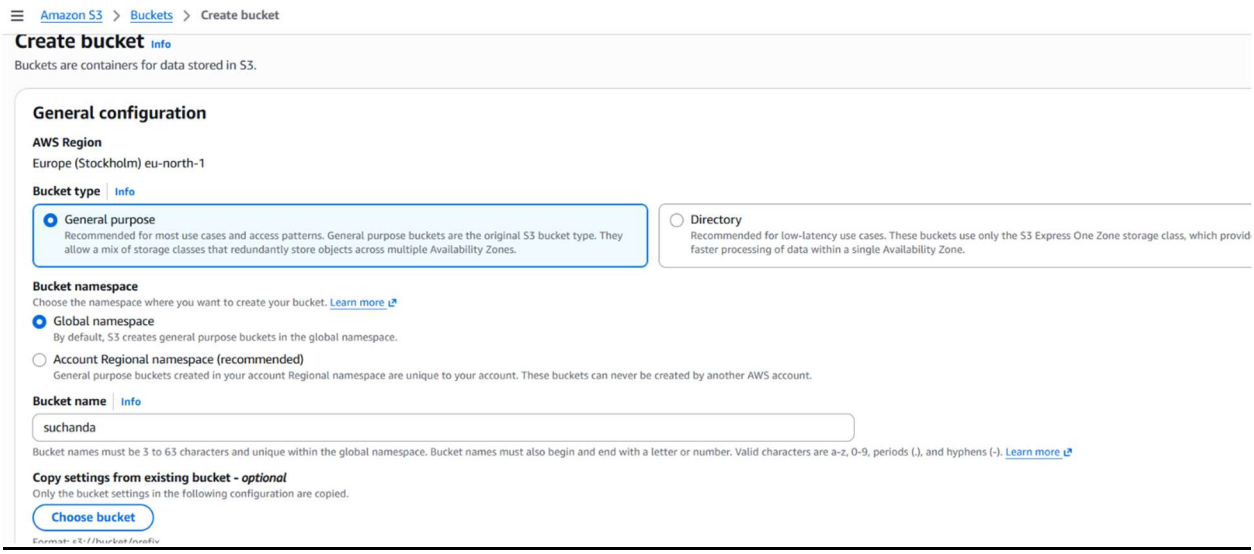
2. Select the AWS Region.

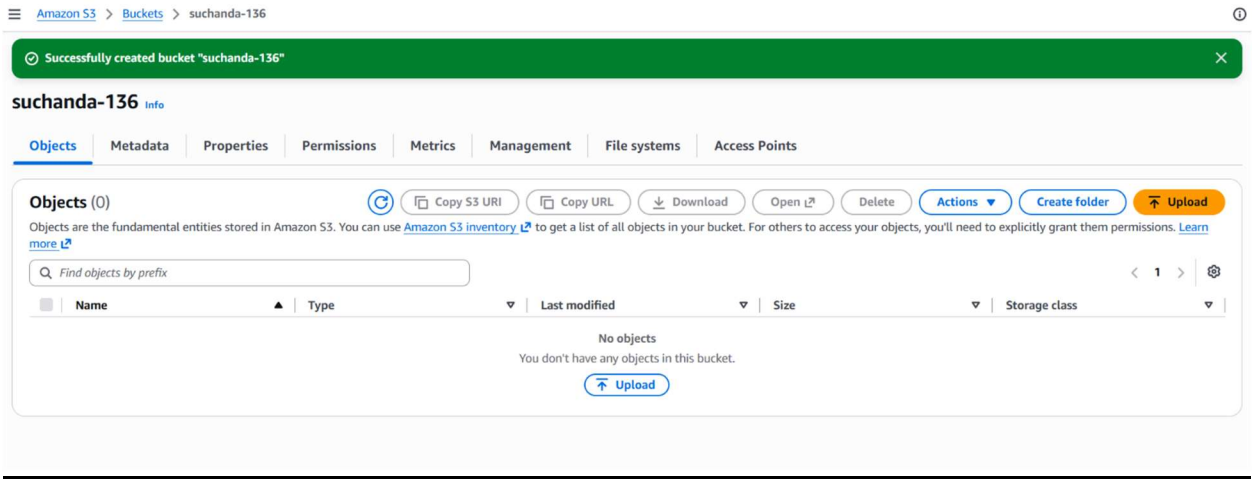
3. Keep Block Public Access enabled.

4. Keep default encryption enabled.

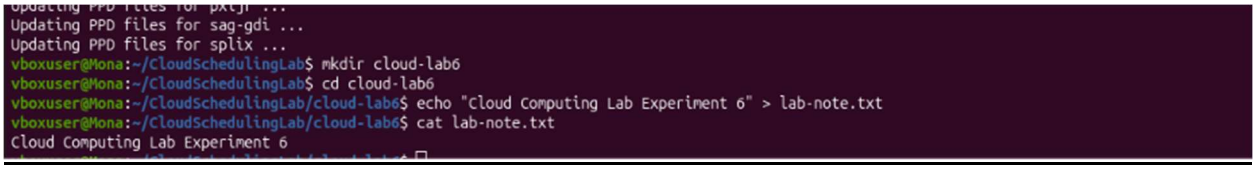
5. Select Create bucket.



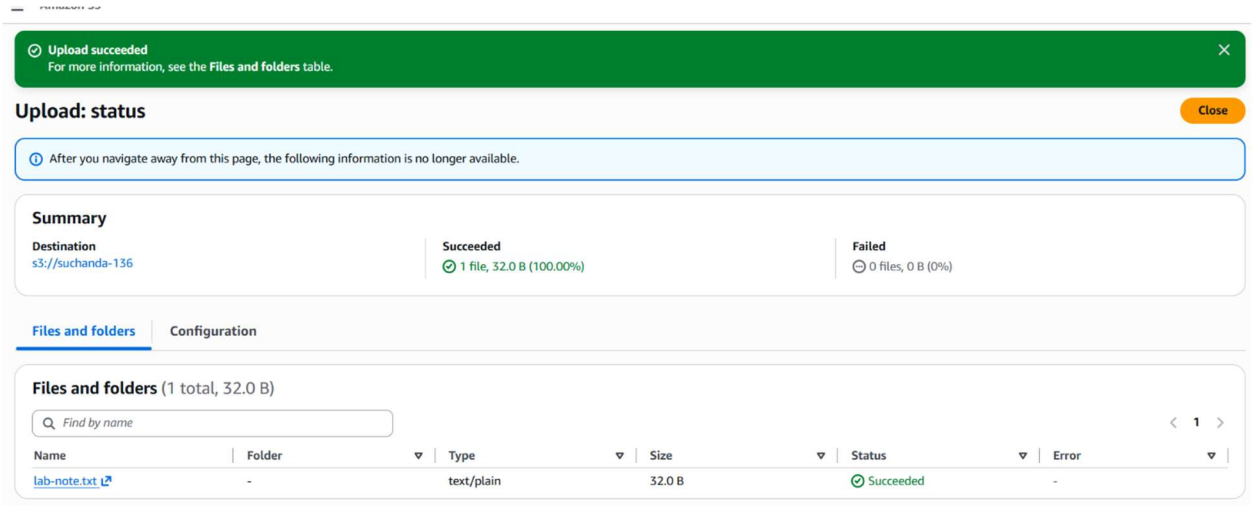




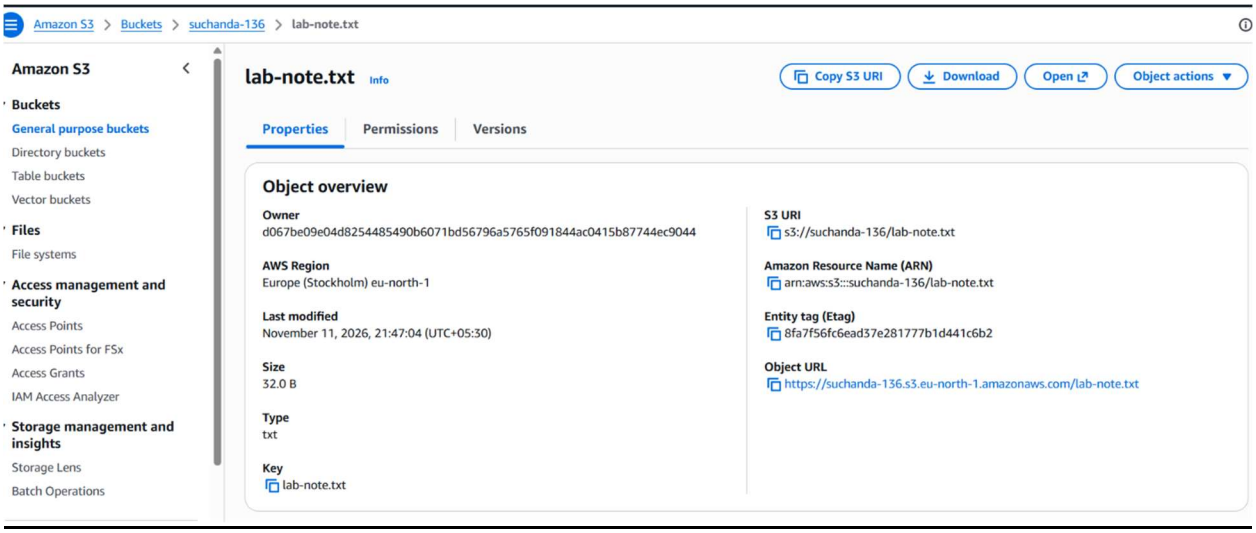
Step 3: Create a Sample File in Ubuntu:



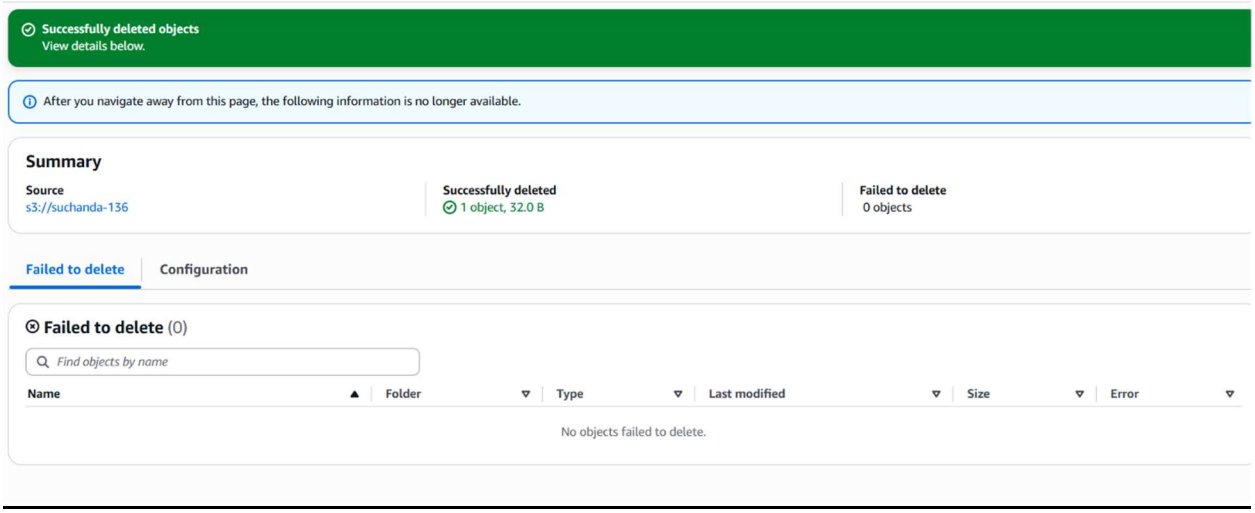
Step 4: Upload a File to S3:

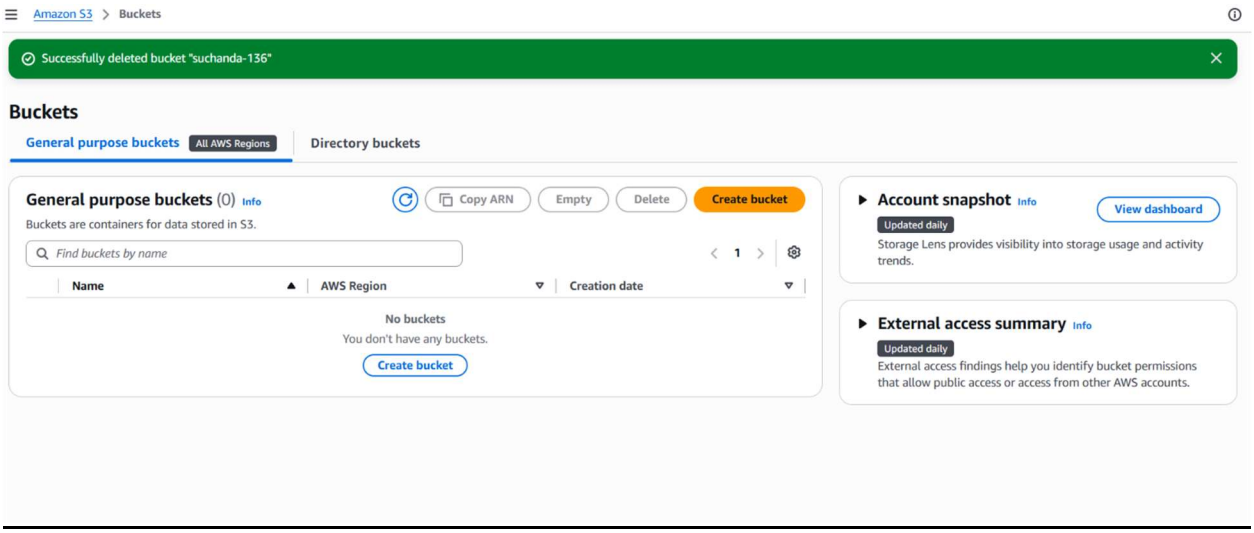


Step 5: Download the File:



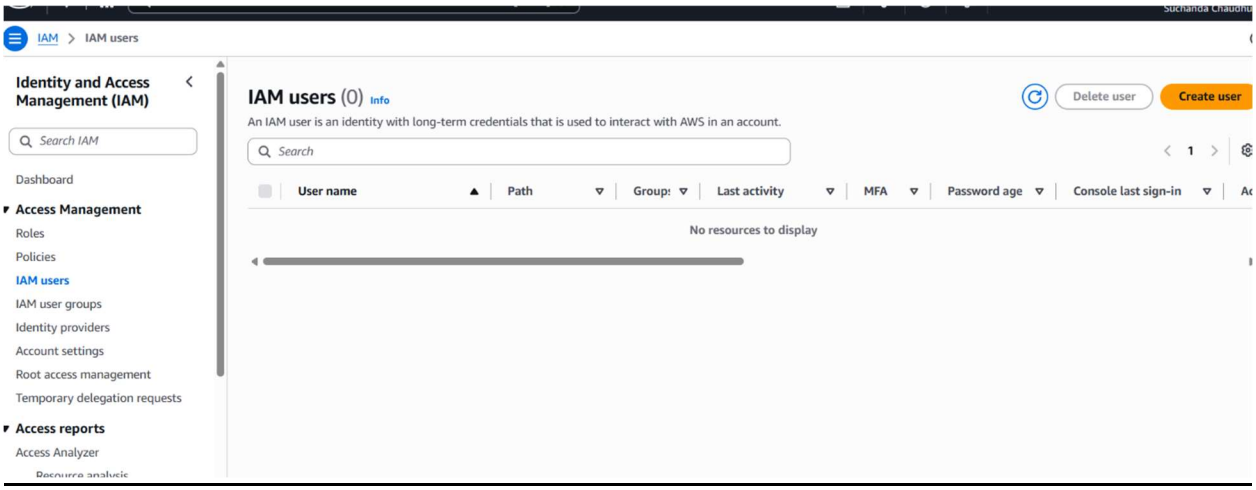
Step 6: Delete the Test Object and Bucket:



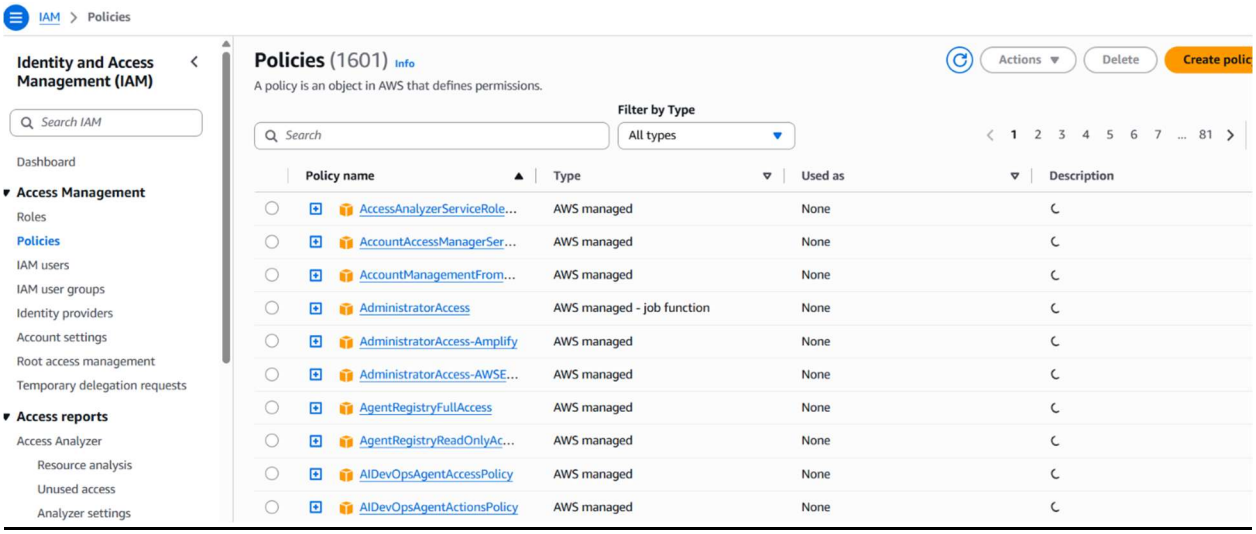


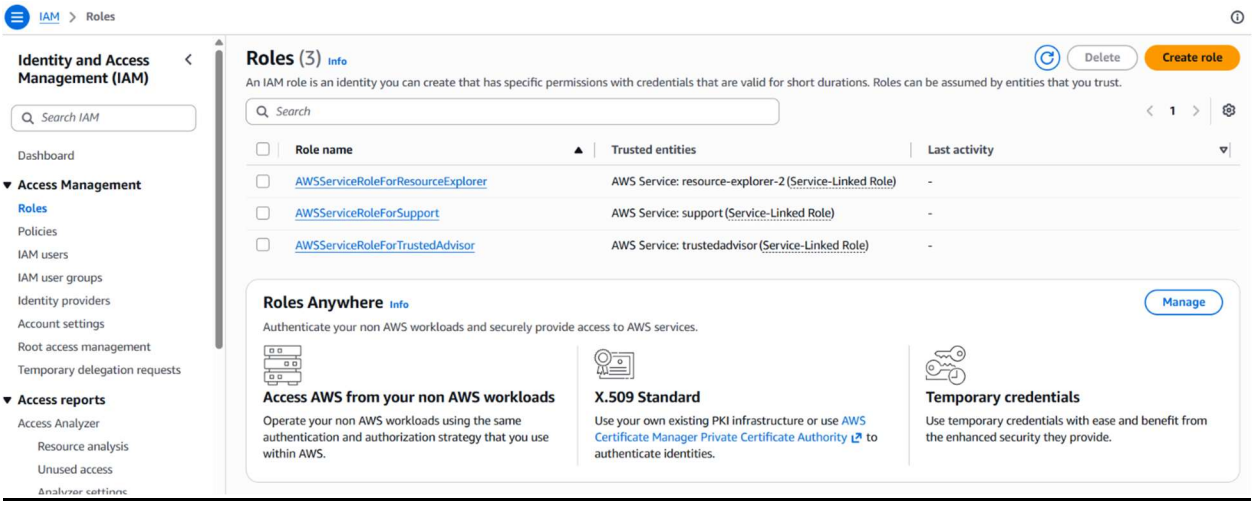
Task 3A: Identity and Access Management Using AWS IAM:

Step 1: Open IAM:



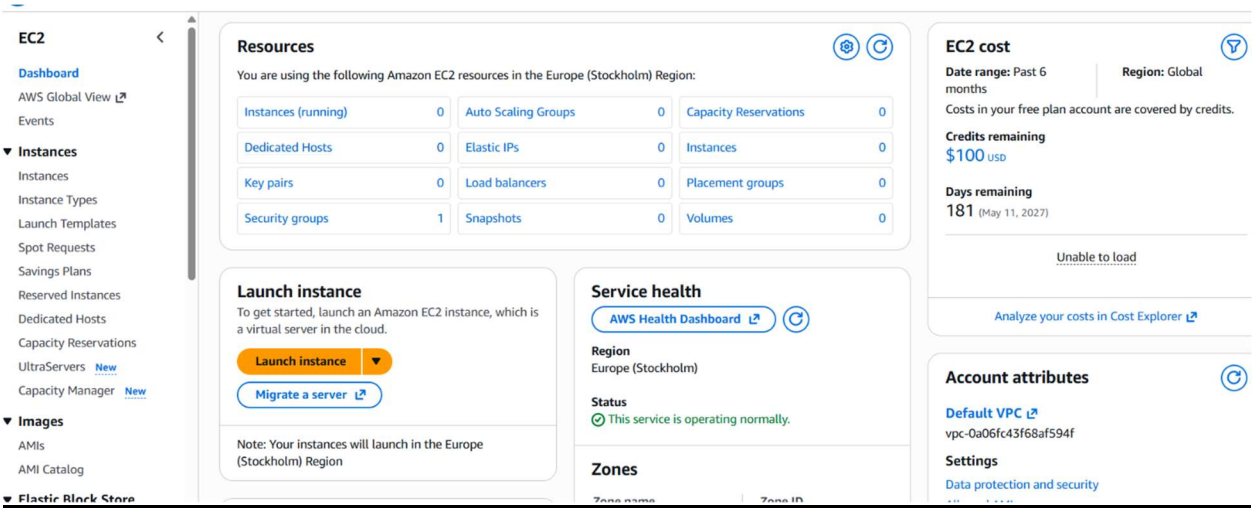
Step 2: Understand IAM Policies:



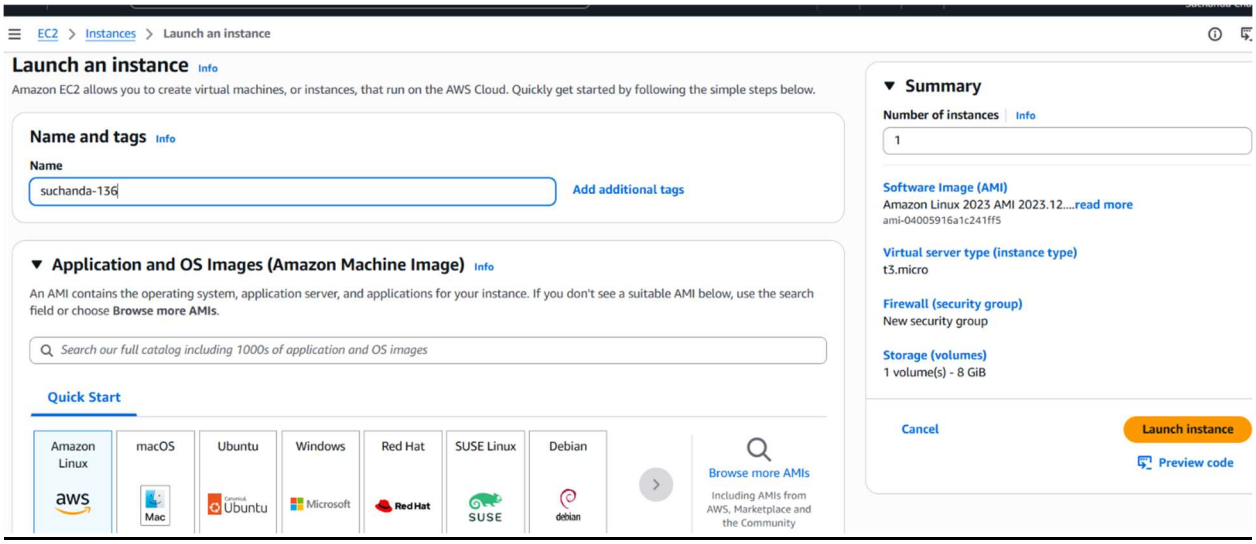


Task 3B: Virtual Machine Provisioning Using Amazon EC2:

Step 1: Open EC2:

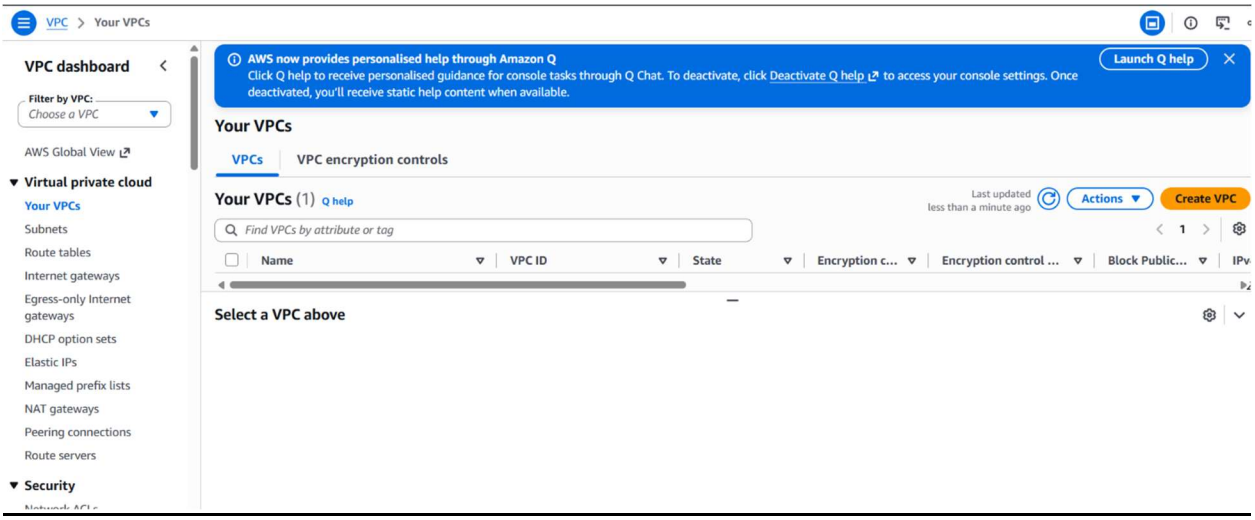


Step 2: Explore the Instance Launch Process:

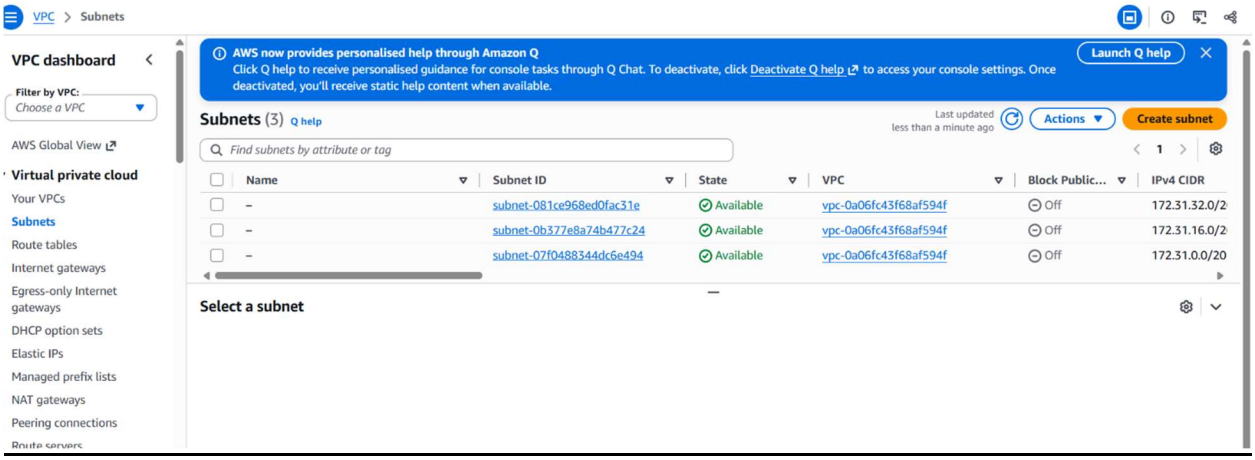


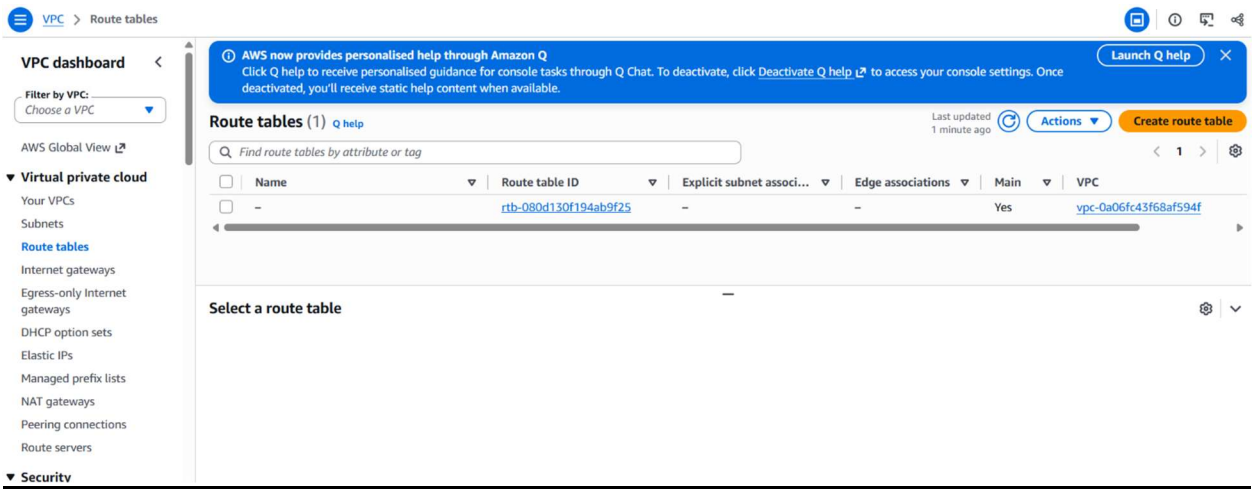
Task 4A: Virtual Networking Using Amazon VPC:

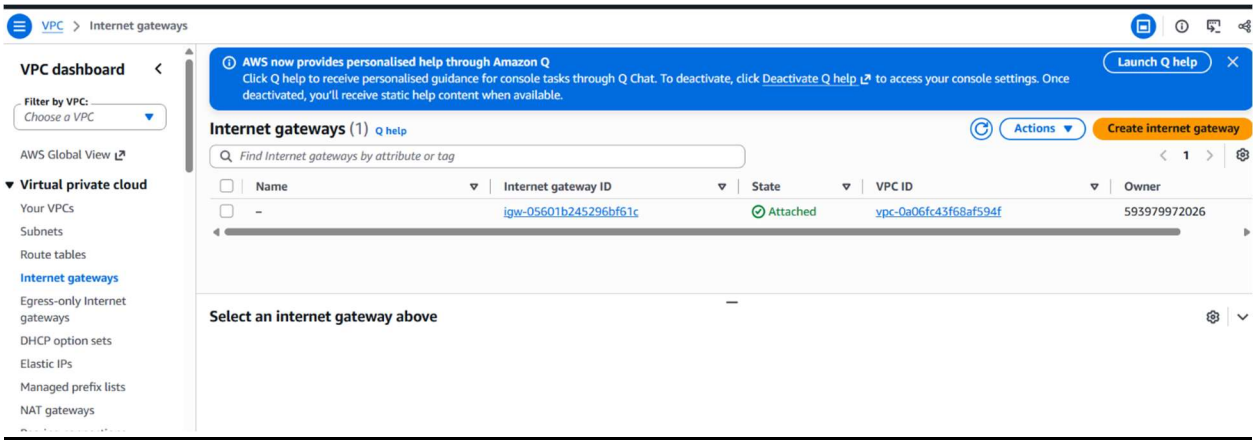
Step 1: Open Amazon VPC:

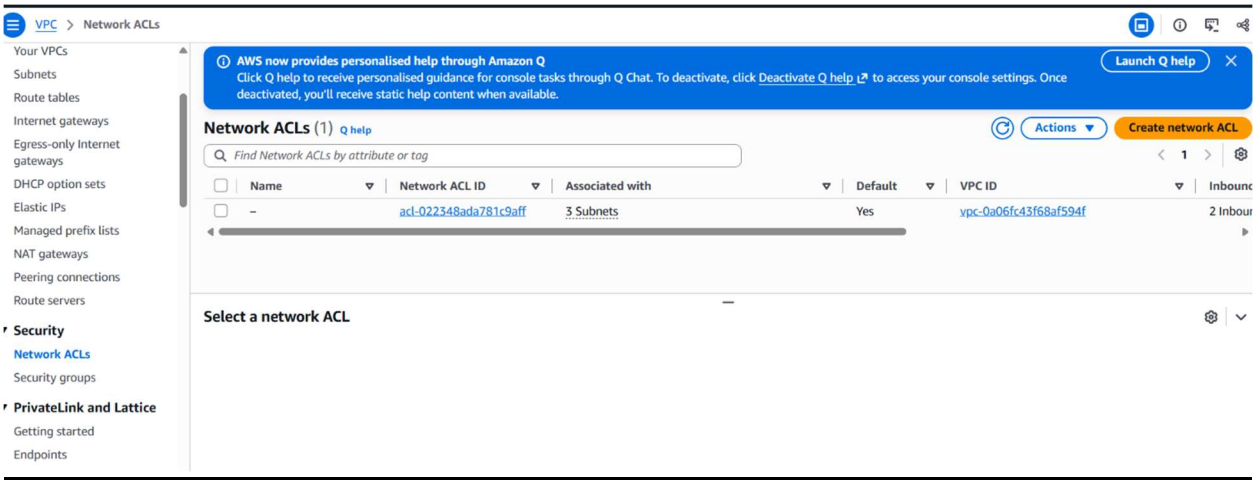


Step 2: Explore VPC Components:



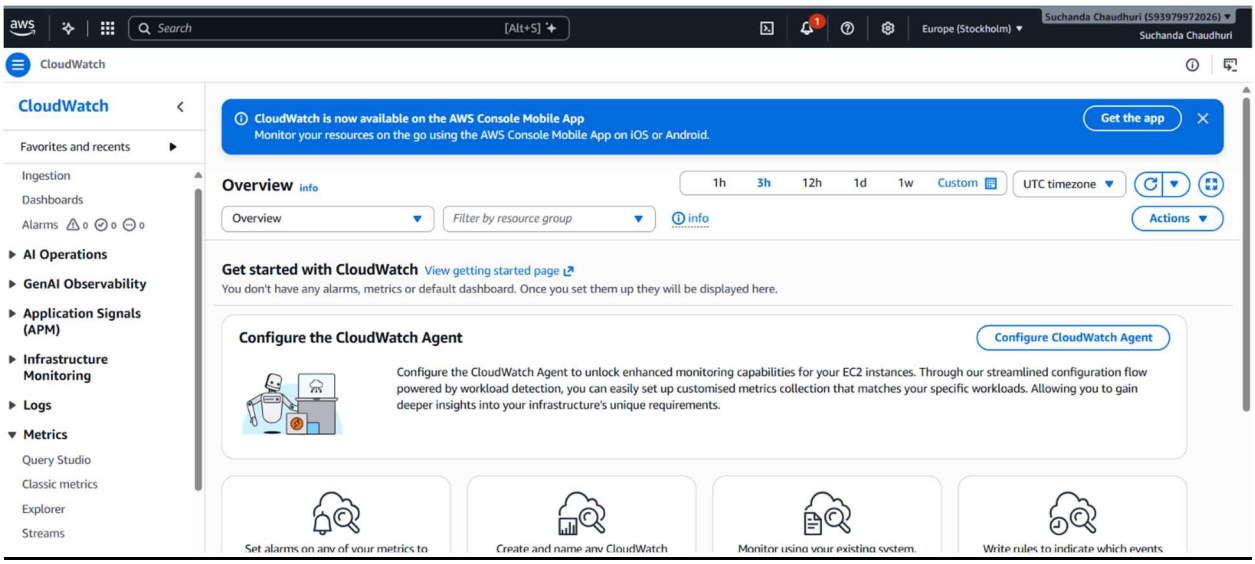

VPC > Route tables
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
AWS now provides personalised help through Amazon Q
Click Q help to receive personalised guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Route tables (1) Q help
Last updated 1 minute ago
Actions
Create route table
Find route tables by attribute or tag
Name
Route table ID
Explicit subnet associ...
Edge associations
Main
VPC
rtb-080d130f194ab9f25
Yes
vpc-0a06fc43f68af594f
Select a route table


VPC > Internet gateways
VPC dashboard
Filter by VPC:
Choose a VPC
AWS Global View
Virtual private cloud
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
AWS now provides personalised help through Amazon Q
Click Q help to receive personalised guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Internet gateways (1) Q help
Actions
Create internet gateway
Find Internet gateways by attribute or tag
Name
Internet gateway ID
State
VPC ID
Owner
igw-05601b245296bf61c
Attached
vpc-0a06fc43f68af594f
593979972026
Select an internet gateway above

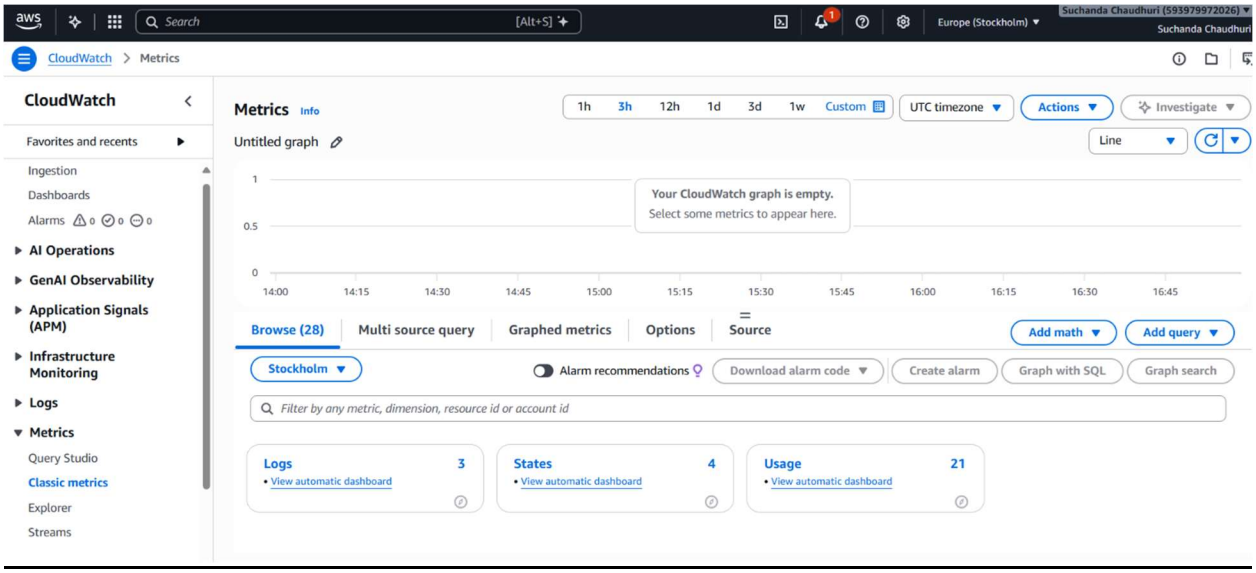

VPC > Network ACLs
Your VPCs
Subnets
Route tables
Internet gateways
Egress-only Internet gateways
DHCP option sets
Elastic IPs
Managed prefix lists
NAT gateways
Peering connections
Route servers
Security
Network ACLs
Security groups
PrivateLink and Lattice
Getting started
Endpoints
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Click Q help to receive personalised guidance for console tasks through Q Chat. To deactivate, click Deactivate Q help to access your console settings. Once deactivated, you'll receive static help content when available.
Launch Q help
Network ACLs (1) Q help
Actions
Create network ACL
Find Network ACLs by attribute or tag
Name
Network ACL ID
Associated with
Default
VPC ID
Inbound
acl-022348ada781c9aff
3 Subnets
Yes
vpc-0a06fc43f68af594f
2 Inbou
Select a network ACL

Task 3D: Cloud Monitoring Using Amazon CloudWatch:

Step 1: Open CloudWatch:



Step 2: Explore Metrics:



Step 3: Explore CloudWatch Alarms:

